

Medical Assistant Project

Python Foundations

UT Austin AI/ML Business Applications Course

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Executive Summary - Actionable Insights

RAG Adds Clear Value Over Plain LLM

Baseline LLM responses (without retrieval) were generic and sometimes hallucinated.

With retrieval and system prompt, answers became more **factual, context-aware, and aligned with the manual.**

Retriever Configuration Impacts Quality

Pure similarity search occasionally surfaced irrelevant chunks (e.g., sepsis query pulled off-topic content).

Thus, retrieval hyperparameters are a critical business lever for accuracy.

Groundedness & Relevance Raters Are Effective

The evaluation layer flagged when an answer drifted from context.

This adds an **explainability/safety mechanism**, useful in regulated industries such as healthcare.

Need to be careful with tokens

With a 2300-token window, long contexts triggered truncation errors.

Careful **chunk sizing (~900/150) and context trimming** balances coverage and cost.

Executive Summary – Business Recommendations

Operationalize RAG for Domain Support - Deploy the pipeline as a domain-specific assistant (e.g., “Medical Manual Assistant”). This improves customer support, staff training, and knowledge management.

Invest in Evaluation Layers - Integrate groundedness/relevance scoring into production. This provides audit trails and builds user trust (“answer sourced from page X”). Strong differentiator for industries with compliance requirements.

Prioritize Retriever Tuning - Standardize on retrieval with $k=5$ as a default baseline. Run A/B tests with k and `chunk_size` to maximize precision/recall balance. Higher accuracy results in fewer escalations, and thus reduces support costs.

Plan for Context Window & Cost Constraints - For long manuals or multi-document corpora, explore hierarchical chunking (sections -> subsections -> sentences). This controls compute spend while maintaining quality.

Domain Expansion – Can suggest extending beyond medical manuals: apply to policy docs, training handbooks, SOPs, compliance regulations. This could reduce employee onboarding time, cut compliance errors, and enhance decision support.

Business Problem Overview and Solution Approach

Problem Statement –

Healthcare professionals are increasingly challenged by **information overload** due to the rapid growth of medical data, research publications, and clinical guidelines. This makes it difficult to quickly locate, interpret, and apply the most relevant and **up-to-date medical knowledge** needed for accurate diagnoses and treatment planning.

The problem is further amplified in **time-sensitive scenarios** such as critical care, where delays in retrieving reliable information can directly impact patient outcomes. Existing manual and keyword-based search methods are inefficient, leading to **decision-making bottlenecks, variability in care practices, and reduced clinical efficiency**.

Thus, there is a pressing need for a solution that can:

- **Streamline access** to trusted medical resources (manuals, guidelines, research papers).
- **Support quick decision-making** by answering domain-specific clinical questions in real time.
- **Ensure accuracy and consistency** by grounding responses in authoritative medical references.
- **Standardize care practices** while improving efficiency in both routine and emergency settings.

Business Problem Overview and Solution Approach

Solution Approach –

This methodology combines **document evaluation -> chunking -> embeddings -> vector retrieval -> LLM generation with prompt engineering -> evaluation guardrails** to create a reliable RAG-based medical assistant that enhances clinical decision-making.

Question Answering using LLM

○ **Model name** = "TheBloke/Mistral-7B-Instruct-v0.2-GGUF"

○ What treatment options are available for managing hypertension?

LLM Response: Hypertension, or high blood pressure, is a common condition that can increase the risk of various health problems such as heart disease, stroke, and kidney damage. The good news is that there are several effective treatment options available to help manage hypertension and reduce the risk of complications. Here are some of the most commonly used treatments:

1. Lifestyle modifications: Making lifestyle changes is often the first line of defense against hypertension. This may include eating a healthy diet rich in fruits, vegetables, whole grains, and lean proteins; limiting sodium intake; getting regular physical activity

Question Answering using LLM

○ Query 1:

○ LLM Response:

Sepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care unit:

1. Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatment as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis.
2. Resusc

Question Answering using LLM

○ Query 2:

- Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis can vary from person to person, but some common signs include:

1. Abdominal pain: The pain may start as a mild discomfort around the navel or in the lower right abdomen, which then gradually moves to the right lower quadrant and becomes more severe over time. The pain may be constant or intermittent and is often worsened by movement, coughing, or deep breathing

Question Answering using LLM

○ Query 3:

- Sudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patches on the scalp, but it can also occur on other parts of the body such as the beard area, eyebrows, or eyelashes.

The exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections, and certain medications.

Question Answering using LLM

○ Query 4:

- A person who has sustained a physical injury to brain tissue, also known as a traumatic brain injury (TBI), may require various treatments depending on the severity and location of the injury. Here are some common treatments recommended for TBIs:

1. Emergency care: The first priority is to ensure the person's airway is clear, they are breathing, and their heart is beating normally. In severe cases, emergency surgery may be required to remove hematomas or other obstructions.
2. Medications: Depending on the symptoms, medications may be prescribed to manage conditions such as

Question Answering using LLM

○ Query 5:

- First and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:

1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.
2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.
3. Immobilize the leg: Use a splint, sl

Question Answering using LLM

- **Observations:**

- The LLM gives coherent, fluent, and medically sensible baseline answers.

However, there are no citations, hallucination risk, lack of grounding in trusted manuals, inconsistent detail, and outdated knowledge.

The LLM alone can simulate medical expertise, but without RAG it cannot guarantee reliability, traceability, or compliance, which are non-negotiable in healthcare.

Question Answering using LLM with Prompt Engineering

- **System prompt** = "You are a medical reference assistant. Answer questions based only on the provided model_path and filename. If the answer is not found in the manual, say you don't know instead of guessing."
- **Query 1:**
- I cannot provide an answer to that question based solely on the given model_path and filename. The provided information does not contain sufficient details about sepsis management protocols in a critical care unit.

Question Answering using LLM with Prompt Engineering

○ Query 2:

- Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis can vary from person to person, but some common signs include:

1. Abdominal pain: The pain is typically located in the lower right quadrant of the abdomen and may start as a mild discomfort that gradually worsens over time. The pain may be constant or intermittent and may be aggravated by movement, deep breathing, or coughing.
2. Loss of appetite

Question Answering using LLM with Prompt Engineering

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Question Answering using LLM with Prompt Engineering

Prompt engineering observations:

- System prompts helps enforce guardrails - This significantly reduced hallucinations compared to vanilla prompting.
- User templates structure question/context alignment - making the LLM less likely to “ignore” them
- Decoding parameters shape reliability and completeness of responses

Together, these elements transformed a generic LLM into a domain-specific medical assistant that is safer, more relevant, and more useful to clinicians.

Data Preparation for RAG

- Data set used - **medical_diagnosis_manual.pdf**
- Parameters used to split the data (***chunk_overlap, chunk_size***) - chunk_size=900, chunk_overlap=150
- Embedding model used - **SentenceTransformerEmbeddings(model_name="all-MiniLM-L6-v2")**
- Parameters used to define the RAG (***k, max_tokens, temperature***) - k = 3, max_tokens=512, temperature=0.2

Question Answering using RAG

○ System prompt -

qna_system_message = "You are a helpful medical reference assistant. Use only the provided context from the medical_diagnosis_manual.pdf to answer questions. If the answer is not in the document, say 'I don't know' instead of guessing. Provide clear, accurate, and concise explanations."

○ User prompt –

qna_user_message_template = "Answer the following question using the provided context:\n\nQuestion: {question}\n\nContext:\n{context}"

Question Answering using RAG

○ Query 1:

- the amount that could be disseminated.

Prevention and Control

The best way to prevent or control an outbreak of biological warfare is to prevent it from occurring. This can be accomplished by international cooperation and diplomacy, as well as by strengthening national security measures. In the event of a suspected attack, the following steps should be taken:

1. Identify the agent (or agents) involved. This may require laboratory analysis of samples obtained from victims or from

Question Answering using RAG

○ Query 2:

- the amount that could be disseminated.

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Question Answering using RAG

- Best parameters for all the questions in the fine-tuning section - temperature=0.5, k=3,max_tokens=200

- Observations -
Improved Accuracy - With RAG, responses were **context-anchored** and more trustworthy

Retriever Behavior Matters - Pure similarity search occasionally returned **off-topic chunks** (seen in the sepsis protocol).

Groundedness and Relevance Raters Add Value - The evaluation layer consistently caught when an answer lacked sufficient support.

Context Window Constraints - The notebook hit token limits (2,300 window). Optimal chunking (chunk_size = 900, overlap = 150) and context trimming are required to balance **coverage vs. model budget**.

Ambiguity in Broad Queries - Vague queries (e.g., “endocrine disorders”) produced **partial or scattered** answers because retrieval spanned multiple sub-sections. More precise queries yielded much better results.

Question Answering using RAG – Fine Tuning

○ Query 1:

- the amount that could be produced. Anthrax is generally considered unlikely to cause widespread panic because it usually produces symptoms only in those who have inhaled large amounts of spores and are exposed for several hours or days. In addition, anthrax can be treated with antibiotics if diagnosed early. Botulinum toxin is a neurotoxin that causes flaccid paralysis by inhibiting the release of acetylcholine from mot

Question Answering using RAG – Fine Tuning

○ Query 2:

- In patients with atypical or equivocal findings, imaging studies should be done without delay. Contrast-enhanced CT has reasonable accuracy in diagnosing appendicitis and can also reveal other causes of an acute abdomen. Graded compression ultrasound can usually be done quickly and uses no radiation (of particular concern in children); however, it is occasionally limited by the presence of bowel gas and is less useful for recognizing nonappendiceal causes of pain. Appendicitis remains primarily a clinical diagnosis. Selective and judicious use of radiographic studies

Question Answering using RAG – Fine Tuning

○ Query 3:

- corticosteroids, retinoids, or immunosuppressants.

Hair loss due to chemotherapy is temporary and is best treated with a wig; when hair regrows, it may be

different in color and texture from the original hair. Hair loss due to telogen effluvium or anagen effluvium

is usually temporary as well and abates after the precipitating agent is eliminated.

Answer:

Sudden patchy hair loss, commonly seen as localized bald spots on the scalp, could be a symptom of alopecia areata

Question Answering using RAG – Fine Tuning

○ Query 4:

- Answer: For a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function, early intervention by rehabilitation specialists is essential for maximal functional recovery. Treatments include prevention of secondary disabilities such as pressure ulcers, joint contractures, and pneumonia, as well as cognitive therapy for severe cognitive dysfunction. Physical and occupational therapy may modestly improve functioning but is more often useful for making the environment safer and providing devices to help patients circumvent primary deficits. The specific rehabilitation therapy varies depending on the patient's abnormalities,

Question Answering using RAG – Fine Tuning

○ Query 5:

- thigh sleeve. Elevation above the level of the heart decreases edema and pain.

Answer: For a person who has fractured their leg during a hiking trip, necessary precautions include preventing Deep Vein Thrombosis (DVT) by encouraging them to walk or move their legs periodically if they are at higher risk. Treatment steps involve rest, ice, compression, and elevation of the affected leg using a thigh sleeve. Pain can be managed with NSAIDs and analgesics as necessary, and crutches may be required initially for walking. For severe injuries,

Question Answering using RAG – Fine Tuning

- **Fine-tuning not strictly required at first** - The base RAG pipeline, with careful prompt engineering, already produced **highly grounded answers** for most queries. Many observed errors were due to **retrieval drift** or **incomplete sources**, not model incapacity.
- **Where fine-tuning helps** - Domain-specific embeddings (trained on medical specific terminology) could improve semantic matching.

Instruction-tuning can also make the LLM answer questions in a more **structured, protocol-like style**.

Groundedness and relevance raters could be fine-tuned to the medical domain, reducing false positives/negatives.

Output Evaluation

○ **Evaluation prompt for groundedness -**

groundedness_rater_system_message = (

"You are an evaluator. Your task is to rate how well the assistant's answer is grounded in the provided context. If the answer is fully supported by the context, mark it as grounded. If parts of the answer are unsupported, speculative, or contradictory, mark it as not grounded. Do not add extra details - just evaluate groundedness."

)

○ **Evaluation prompt for relevance -**

relevance_rater_system_message = (

"You are an evaluator. Your task is to judge how relevant the retrieved context is to the user's question. If the context directly supports answering the question, rate it as relevant. If it is only loosely connected, off-topic, or irrelevant, rate it as not relevant. Do not invent new information."

)

Output Evaluation

- **Query 1:** ground, Based on the user's question about the protocol for managing
- **Query 2:** The answer is grounded as it accurately summarizes the common symptoms of appendicitis from the context and correctly states that it cannot be
The context is highly relevant to the user's question as it provides detailed information about the common symptoms of appendicitis and confirms that treatment for appendicitis involves surgical removal
- **Query 3:** The assistant's answer is grounded in the context as it correctly identifies alopecia areata as a cause of sudden patchy hair loss and mentions various treatment options such as topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA) that are consistent with the context. Relevant. The context provides detailed information about the causes and treatments of alopecia areata, which is a type of sudden patchy hair loss that matches the user's question.
- **Query 4:** Grounded. The answer accurately summarizes the treatments recommended for a person with a brain injury based on the information provided in the context. Relevant. The context provides detailed information about the recommended treatments for a person with a brain injury, including early intervention by rehabilitation specialists, prevention of secondary disabilities, physical and occupational therapy, cognitive therapy, and family education.

Output Evaluation

○ Query 5:

The answer is grounded in the context provided. It correctly identifies the risk of DVT for someone with a fractured leg and recommends preventive measures based on the information given about different treatment options for DVT prevention. Additionally, it mentions the initial care steps of rest, ice, and compression, which are also mentioned in the context. (NSAIDs) and analgesics may be prescribed for pain relief, and crutches might be necessary if walking is painful.

3. Further treatment: Depending on the severity of the injury, splinting or casting might be required to prevent further injury and decrease pain. In some cases, definitive treatment like reduction (realignment of the bone) may be necessary, which usually requires analgesia or sedation. Closed reduction is maintained by casting, while open reduction is maintained by surgical hardware such as pins, screws, plates, or external fixators.

4. Prevention and recognition of complications: It's essential to recognize and prevent potential complications like nerve injuries, arterial injuries, or infections. Nerve conduction studies may be necessary for nerve injuries, while arteriography might be required for suspected arterial injuries. Treatment for life-threatening injuries includes surgical

Output Evaluation

Question	Relevance (1-5)	Groundedness	Comments / Observations
Pulmonary embolism - symptoms & treatments	4	Grounded	Retrieval tends to surface symptoms (dyspnea, pleuritic chest pain, tachycardia) and management (anticoagulation, thrombolysis criteria). Symptoms and treatments may be split across sections.
Hypertension - trade names of medications	2	Partial	Many manuals emphasize generic names. Trade names may be sparse or absent and model may generalize (risk of hallucination).
Rheumatoid arthritis - first-line & alternatives	4	Grounded	Retrieval reliably surfaces DMARDs (methotrexate first-line), escalation (leflunomide, sulfasalazine), biologics/JAK inhibitors, monitoring/contraindications
Endocrine disorders - diagnostic steps	3	Partial	"Endocrine disorders" is broad; retrieval can scatter across thyroid/adrenal/pituitary. Answers may be general
Sepsis - ICU protocol	3	Partial -> Grounded (after tuning)	When the correct chapter is retrieved, answers match up well

Files used –

- medical_diagnosis_manual.pdf
- Low_Code_NLP_RAG_Project_Notebook.ipynb

APPENDIX

Data Background and Contents

The **Merck Manuals** are medical references published by the American pharmaceutical company Merck & Co., that cover a wide range of medical topics, including disorders, tests, diagnoses, and drugs. The manuals have been published since 1899, when Merck & Co. was still a subsidiary of the German company Merck.

The manual is provided as a PDF with over 4,000 pages divided into 23 sections.



Power Ahead!

